Toqa Mahmoud Tawfiq Al-zoubi (32209303103)

Shaden Khaled Mohammad Alshobake (32209303017)

Ghaida Abd-Almajied Yousef Breitem (32209303009)

Supervised Name : Prof. Abdelwadood MESLEH

Part 1: Project Overview

Title: Graph-Based Text Mining for Research Paper Classification.

Dataset: The Cora Citation Network.

Objective:

The objective of this project is to develop a deep learning model capable of classifying research papers into one of seven academic categories by integrating both textual content and citation relationships. Research papers usually contain important textual information such as keywords, topics, and technical terms. However, the content of a paper alone may not always be enough to accurately identify its research field. For this reason, the project also considers the citation relationships between papers.

In this project, the Cora dataset is represented as a graph, where each research paper is treated as a node, and each citation link between two papers is treated as an edge. This graph-based representation allows the model to learn not only from the features of each individual paper, but also from the surrounding papers connected to it through citations.

The main idea is that papers connected by citation links often share related topics or belong to similar research areas. Therefore, by combining text mining with graph learning, the model can capture richer information and improve classification performance. This makes the project a practical example of using Graph Neural Networks for structured text mining and research paper classification.

Part 2: Methodology

Intelligent Text Mining

The Cora dataset represents each research paper using a bag-of-words feature vector. Each paper contains **1433 text-based features**, which describe the presence or absence of important words in the document.

Link Analysis and Graph Learning

Instead of treating research papers as independent samples, the papers were represented as nodes in a graph. Citation relationships between papers were treated as edges. This allowed the model to learn not only from the content of each paper, but also from its neighboring papers in the citation network.

Graph Neural Networks

Several graph-based neural network models were implemented and compared:

1. **GCN — Graph Convolutional Network**
   Used message passing and neighborhood aggregation to classify papers based on their features and citation links.
2. **GAT — Graph Attention Network**
   Improved the learning process by using attention mechanisms, allowing the model to assign different importance weights to neighboring papers.
3. **GATv2 — Graph Attention Network Version 2**
   Tested as an advanced attention-based variant to examine whether a more flexible attention mechanism could improve classification performance.

The final selected model was the **Improved GAT model**, because it achieved the highest test accuracy.

Part 3: System Architecture

Input Layer

The input layer receives the Cora feature matrix with the following shape:

```
2708 × 1433
```

This means the dataset contains **2708 research papers**, and each paper is represented by **1433 text features**.

Graph Structure

The citation network contains:

```
10556 edges
```

These edges represent citation relationships between research papers.

Improved GAT Architecture

The best-performing model was based on a Graph Attention Network.

First GAT Layer

The first GAT layer applies multi-head attention to learn richer node representations. It uses:

```
8 attention heads
8 hidden features per head
Dropout = 0.5
ELU activation
```

This layer allows each paper to learn from its neighboring papers while giving different importance to each neighbor.

Second GAT Layer

The second GAT layer maps the learned hidden representation to the final output classes:

```
7 output classes
```

Output Layer

The output layer produces class scores for each paper. The predicted class is selected based on the highest probability score.

Part 4: Discussion of Results

Several experiments were conducted to evaluate and improve the classification performance of the proposed graph-based text mining system. The main goal of these experiments was to compare different Graph Neural Network architectures and determine which model achieved the best generalization on unseen research papers.

4.1 Experimental Comparison

Experiment	Model	Main Improvement	Test Accuracy
Baseline GCN	GCN	Basic graph convolution and neighborhood aggregation	80.50%
Baseline GAT	GAT	Attention-based neighborhood learning	79.85%
Improved GAT	GAT	Early stopping, validation-based model selection, and learning rate scheduling	**82.50%**
GATv2 Multi-Seed	GATv2	Advanced attention mechanism with multiple seed testing	82.10%

The experimental results show that the **Improved GAT model** achieved the highest test accuracy among all tested models. Although the baseline GCN achieved a strong result of **80.50%**, the improved GAT model performed better by using attention mechanisms and training optimization techniques.

4.2 Final Accuracy

The best-performing model was the **Improved GAT model**, which achieved:

```
Final Test Accuracy: 82.50%
```

This result improved the previous baseline GAT accuracy of **79.85%** by approximately:

```
2.65%
```

It also outperformed the baseline GCN model, which achieved **80.50%** test accuracy. This indicates that the attention-based model, when combined with proper training strategies, was more effective for this graph-based classification task.

4.3 Improved GAT Result

The improved GAT model achieved the following results:

Metric	Result
Best Epoch	16
Final Train Accuracy	97.86%
Final Validation Accuracy	80.80%
Final Test Accuracy	**82.50%**

The best model was selected based on **validation accuracy** rather than simply using the final training epoch. This approach helped reduce the effect of overfitting and improved the model's ability to generalize to unseen test nodes.

The improvement came mainly from using:

Technique	Purpose
Early Stopping	Stops training when validation performance no longer improves
Validation-Based Model Selection	Saves the best model according to validation accuracy
Learning Rate Scheduler	Adjusts the learning rate during training
Dropout Regularization	Reduces overfitting
Weight Decay	Penalizes overly complex model weights

4.4 GATv2 Multi-Seed Experiment

To further investigate whether a more advanced attention mechanism could improve the results, a **GATv2** model was tested using multiple random seeds. This experiment was important because graph neural networks can be sensitive to random initialization, dropout, and training randomness.

Seed	Best Epoch	Train Accuracy	Validation Accuracy	Test Accuracy
42	17	100.00%	80.00%	81.20%
7	8	98.57%	79.40%	79.80%
123	16	99.29%	81.20%	**82.10%**
2024	8	97.14%	76.40%	78.30%
2026	5	96.43%	78.60%	79.30%

The best GATv2 result was obtained using seed **123**, with the following performance:

Metric	Result
Best Seed	123
Best Validation Accuracy	81.20%
Best Test Accuracy	82.10%

Although GATv2 achieved strong performance, it did not outperform the improved GAT model. Therefore, GATv2 was considered an additional advanced comparison experiment rather than the final selected model.

4.5 Convergence and Training Behavior

The training results showed that the models were able to learn the training data effectively. In several experiments, the training accuracy reached very high values, sometimes close to **100%**. However, the validation and test accuracy remained around **80–82%**, which indicates that some overfitting occurred.

This behavior is expected in citation network datasets such as Cora, where the number of labeled training nodes is relatively limited compared to the total number of papers. To reduce this issue, the final improved model used early stopping, dropout regularization, weight decay, validation-based model selection, and learning rate scheduling.

These techniques helped select the best-performing model before the model became too specialized to the training nodes.

4.6 Scientific Conclusion

The results confirm that graph-based learning is effective for research paper classification. Citation links provide important relational information that text features alone may not capture. By using graph neural networks, the model was able to combine textual features with citation relationships to improve classification performance.

The improved GAT model achieved the best performance because it used attention mechanisms to assign different importance levels to neighboring papers. This allowed the model to focus more on the most relevant citation relationships instead of treating all neighboring papers equally.

Therefore, the project demonstrates the effectiveness of **Graph Neural Networks** and **Relational Deep Learning** in structured text mining tasks, especially when both document content and citation relationships are available.

Convergence and Training Behavior

The training results showed that the model learned the training data effectively. In several experiments, the training accuracy reached a very high value, sometimes close to 100%. However, the validation and test accuracy remained around 80–82%, which indicates that some overfitting occurred.

To handle this issue, the final improved model used:

```
Early Stopping
Validation-based model selection
Learning rate scheduling
Dropout regularization
Weight decay
```

These techniques helped select the best model before overfitting became stronger.

Scientific Conclusion

The results confirm that graph-based learning is highly effective for research paper classification. Citation links provide important relational information that text features alone may not capture.

The improved GAT model achieved the best performance because it used attention mechanisms to assign different importance levels to neighboring papers. This allowed the model to focus more on the most relevant citation relationships.

Therefore, the project demonstrates the effectiveness of **Graph Neural Networks** and **Relational Deep Learning** in structured text mining tasks.

Part 5: Execution Steps

1. **Environment Setup**
   Ensure Python 3.10 is installed.
2. **Install Dependencies**
   Run the following command:

```
pip install -r requirements.txt
```

This installs the required libraries such as:

```
PyTorch
Torch-Geometric
Pandas
Matplotlib
NumPy
```

3. **Data Placement**
   Ensure that the dataset files are located inside the `/data` directory.
4. **Run the First Baseline Experiment**

```
python main_1.py
```

5. **Run the Second Experiment**

```
python main_2.py
```

6. **Run the Improved GAT Model**

```
python main_3.py
```

This experiment produced the best result:

```
Final Test Accuracy: 82.50%
```

7. **Run the GATv2 Multi-Seed Experiment**

```
python main_4.py
```

This experiment tested several random seeds and selected the best model based on validation accuracy.

8. **Generated Output Files**

The training process generates saved model files and learning curve images, including:

```
best_gat_model.pt
best_gatv2_seed_123.pt
learning_curves_improved.png
learning_curves_gatv2_seed_123.png
```

Final Selected Model

Based on the experimental results, the final selected model is:

```
Improved Graph Attention Network
```

With the best achieved result:

```
Final Test Accuracy: 82.50%
```

This model was selected because it achieved the highest test accuracy among all tested models while using validation-based model selection to improve generalization.